ConvoShield AI – Implementation Plan

GDG Code4Her Hackathon | Track 2: Cybersecurity & Privacy for Women | AI-Based Suspicious Conversation Analyzer

📋 Problem Statement

Online scams, grooming attempts, and manipulative conversations often begin with seemingly normal interactions and gradually escalate. This tool analyzes conversation patterns using AI and identifies potential indicators of scams, manipulation, or harassment.

🏗️ Architecture

Layer	Technology	Rationale
Frontend	Vite + React	Fast setup, hot reload, great DX
Styling	Vanilla CSS + CSS Variables	Full control, dark cybersecurity theme
AI Engine	NVIDIA AI API (OpenAI-compatible)	Powerful LLMs, user has API key
Model	meta/llama-3.1-70b-instruct	Excellent instruction-following
PDF Export	html2pdf.js	Client-side report generation

> ℹ️ **Note:** The app runs entirely in the browser — no backend server required. API calls go directly to NVIDIA's endpoint. The user's API key is stored in localStorage only.

📁 Project Structure

File	Purpose
`src/services/nvidiaService.js`	NVIDIA API integration, prompt engineering, JSON parsing
`src/data/sampleConversations.js`	5 pre-loaded demo conversations (scam, phishing, grooming, job scam, safe)
`src/components/Hero.jsx`	Landing section with animated background & feature highlights
`src/components/ChatInput.jsx`	Conversation input with drag-and-drop, character count
`src/components/AnalysisResult.jsx`	Full results dashboard with risk gauge & tabs

`src/components/RiskGauge.jsx`	SVG animated semicircular risk meter (0–100)
`src/components/FlaggedPhrases.jsx`	Highlighted suspicious phrases with reasons & tactic labels
`src/components/SafetyAdvice.jsx`	Numbered recommendations + emergency helplines
`src/components/SampleConversations.jsx`	One-click demo cards for hackathon presentation
`src/components/ApiKeyModal.jsx`	NVIDIA API key input with privacy info

🤖 AI Service Design

Prompt Engineering

The system prompt instructs the LLM to act as an expert cyber-safety analyst and return structured JSON with:

- **riskLevel:** low / moderate / high
- **riskScore:** 0–100 numeric score
- **conversationType:** scam, grooming, harassment, phishing, manipulation, or safe
- **flaggedPhrases:** Exact suspicious text with reason and tactic name
- **tactics:** List of all detected manipulation tactics
- **safetyAdvice:** Actionable safety recommendations
- **detailedAnalysis:** Paragraph explaining the pattern

API Integration

Uses `fetch()` to call `https://integrate.api.nvidia.com/v1/chat/completions` with OpenAI-compatible format. Model: `meta/llama-3.1-70b-instruct`, Temperature: 0.3 for consistent outputs.

🎨 Design Philosophy

- **Dark cybersecurity aesthetic** — deep navy (#0a0e27), neon cyan/purple gradients
- **Glassmorphism cards** — frosted glass panels with backdrop blur
- **Micro-animations** — floating orbs, scan-line effects, pulse glows
- **Color-coded risk** — green (safe) → amber (moderate) → red (high)
- **Responsive** — works on desktop, tablet, and mobile

🏆 Hackathon-Winning Differentiators

1. Stunning dark cyber-aesthetic UI with glassmorphism & neon gradients
2. Real-time analysis animation with scanning effect during AI processing

3. Detailed tactic breakdown — names specific manipulation techniques
4. PDF report export — downloadable safety report
5. Pre-loaded demo conversations — smooth hackathon presentation
6. Mobile responsive design
7. No backend server — runs entirely in browser

✅ Verification Plan

Test Scenarios

- Romance scam conversation → should detect love bombing, urgency, financial bait
- Financial fraud → should detect authority impersonation, urgency, data harvesting
- Online grooming → should detect flattery, secrecy, gift-giving tactics
- Safe conversation → should return low risk with no false positives

Manual Verification

- Enter NVIDIA API key and verify connection
- Test with real suspicious conversations
- Verify PDF export functionality
- Check responsive layout on mobile

ConvoShield AI — Built for GDG Code4Her Hackathon 2026
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